

What's New for SIGGRAPH Educators in CS2023: Undergraduate Computer Science Curricular Guidelines

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Abstract

The Computer Science Curricula 2023 (CS2023) was endorsed by their sponsoring organizations: the Association of Computing Machinery (ACM), IEEE Computing Society (IEEE CS), and the Association for the Advancement of Artificial Intelligence (AAAI) in early 2024. The document was produced over the course of four years and was the collaborative work of approximately 100 volunteers from six continents. In keeping with current curricular design principles, the guidelines incorporate a competency model (what is learned) in addition to the traditional knowledge model (what is taught). This talk is geared to anyone interested in computer science graphics education who would like to learn more about the guidelines and the on-going computing curricular efforts. We will describe the organization of CS2023, the Graphics and Interactive Techniques Knowledge Area including its new Knowledge Unit, Society Ethics and the Profession.

CCS Concepts

• **Social and professional topics** → **Model curricula**.

Keywords

CS2023, Curricular Guidelines

ACM Reference Format:

Susan Reiser and Jeff Lait. 2024. What's New for SIGGRAPH Educators in CS2023: Undergraduate Computer Science Curricular Guidelines. In *SIGGRAPH Asia 2024 Educator's Forum (SA Educator's Forum '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3680533.3697065>

1 Introduction

Undergraduate curricular guidelines have been published roughly every ten years since 1968. Initially the guidelines were authored by ACM alone. In 1991 IEEE CS began collaborating on their development and, because of the increasing importance of artificial intelligence, AAAI collaborated on the latest version. All three organizations endorsed CS2023 [Kumar et al. 2024] in the first quarter of 2024.

CS2023's body of knowledge or the knowledge model is organized as a collection of seventeen knowledge areas including the

Graphics and Interactive Techniques Knowledge Area. Because of the need for computer science as a discipline to address issues of society and the profession, the guidelines emphasize infusing those topics throughout the undergraduate curriculum including within computer graphics. The body of knowledge is followed by the competency model. The complete guidelines are freely available at <https://dl.acm.org/doi/pdf/10.1145/3664191>.

The Graphics Knowledge Area as CS2023 was a team effort. The graphics sub-committee consisted of Erik Brunvand, University of Utah; Kel Elkins, NASA/GSFC Scientific Visualization Studio; Jeff Lait, SideFX; Amruth Kumar, Ramapo College; Paul Mihail, Valdosta State University; Tabitha Peck, Davidson College; Susan Reiser, UNC Asheville; Ken Schmidt, NOAA NCEI; and Dave Shreiner, Unity Technologies and Sonoma State University. We are indebted to other contributors and reviewers including: Ginger Alford, Southern Methodist University; Christopher Andrews, Middlebury College; A. J. Christensen, NASA/GSFC Scientific Visualization Studio; Roger Eastman, University of Maryland; Ted Kim, Yale University; Barbara Mones, University of Washington; Greg Shirah, NASA/GSFC Scientific Visualization Studio; Beatriz Sousa Santos, University of Aveiro; and Anthony Steed, University College, London.

2 Graphics and Interactive Techniques Knowledge Area

A knowledge area is a set of knowledge units which in turn is comprised of a set of related topics coupled with learning outcomes. SIGGRAPH and SIGGRAPH Asia's common theme provided our impetus to rename CS2013's Graphics and Visualization Knowledge Area to CS2023's Graphics and Interactive Techniques Knowledge Area. This new name better exemplifies the expanded content of the knowledge area and aligns with the conferences' content. Graphics and Interactive Techniques do not stand alone but are an integral part of an institution's computer science curriculum. Some universities may not offer a single graphics course, others may deliver an entire computer graphics track or major. CS2023 is designed to provide flexible scaffolding for course and program design. It is intentionally not prescriptive. Thus, the number of required graphics core hours is only four, yet the provided content spans eleven thematic knowledge units.

3 Society, Ethics and the Profession in Graphics

All three sponsoring associations have codes of ethics and since 1991 [Force 1991], the curricular guidelines have included ethics. CS2023 recommends teaching the knowledge units found in Society, Ethics and the Profession in Graphics throughout the curriculum

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SA Educator's Forum '24, December 03–06, 2024, Tokyo, Japan

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ACM ISBN 979-8-4007-1136-7/24/12

<https://doi.org/10.1145/3680533.3697065>

Table 1: Graphics Related Knowledge Units (listed in document order)

CS2023	CS2013
Fundamental Concepts	Fundamental Concepts
Visualization	Basic Rendering
Applied Rendering and Techniques	Geometric Modeling
Geometric Modeling	Advanced Rendering
Shading and Advanced Rendering	Computer Animation
Computer Animation	Visualization
Simulation	
Immersion	
Interaction	
Image Processing	
Tangible/Physical Computing	
Society, Ethics, and the Profession	

which stresses their importance to the field. This can be accomplished authentically. For example, while discussing images, an

educator can have a short conversation about privacy and intellectual property. While intellectual property law and expectations of privacy vary in different countries, the concepts exist in all. Because of the timely importance of these topics the majority of knowledge areas content related to Society, Ethics and the Profession.

Acknowledgments

This work was made possible by support from the following organizations: Association for Computing Machinery (ACM), IEEE Computer Society (IEEE-CS), Association for the Advancement of Artificial Intelligence (AAAI), and the National Science Foundation.

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